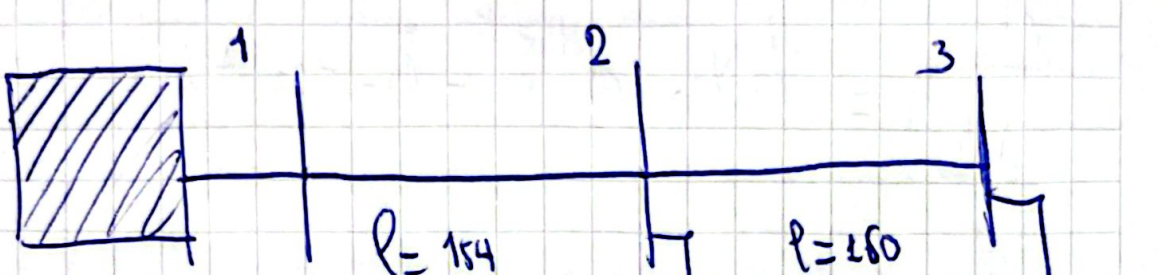
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Mijabbar Badalov / 285269

MPA-RES 26/27 Z

Homework 3:



$V = 250\,kV$

$V_n = 220\,kV$

$S_b = 100\,MVA$

$l = 154$

$R_K = 0$

$X_K = 0{,}31$

$B_K = 2{,}7$

$P = 150$

$Q = 20$

3φ

$l = 250$

$X_K = 0{,}3$

$B_K = 3{,}1$

$P = 95$

$Q_3 = 9$

Base Values:

$$Z_b = \frac{V_b^2}{S_b} = \frac{220^2}{100} = 484\ \Omega \qquad Y_b = \frac{1}{Z_b} = 2{,}0661\,mS$$

Line Parameters:

1–2

$$X_{12} = 0{,}31 \times 154 = 47{,}74\,\Omega$$

$$B_{12} = 2{,}7 \times 10^{-6} \times 154 = 415{,}8\,\mu S$$

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$$x_{12} = \frac{47,74}{484} = 0,09864$$

$$b_{12} = 415,9 \times 10^{-6} \times 484 = 0,20125$$

- 2 – 3:

$$X_{23} = 0,3 \times 160 = 48,00\,\Omega$$

$$B_{23} = 2,7 \times 10^{-6} \times 160 = 432,0\,\mu S$$

$$x_{23} = \frac{48}{484} = 0,09911$$

$$b_{23} = 432,0 \times 10^{-6} \times 484 = 0,20909$$

- Admittance Matrix:

$$Y_{21} = -\frac{1}{jx_{12}} = -\frac{1}{j0,09864} = j10,1382 = 10,1382\angle 90°$$

$$Y_{22} = \frac{1}{jx_{12}} + \frac{1}{jx_{23}} + \frac{jb_{12}}{2} + \frac{jb_{23}}{2} =$$

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$$= -j10,1382 - j10,0833 + j0,10062 + j0,10454 =$$

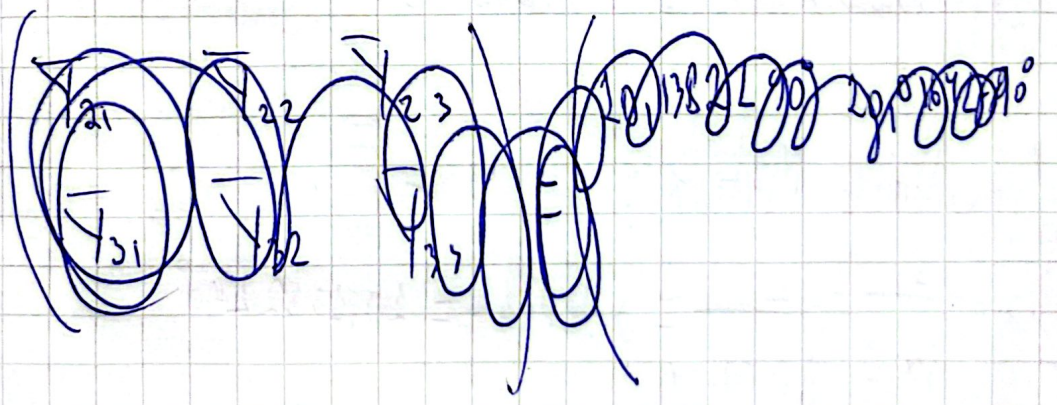
$$= -j20,10164 = 20,10164 \angle -90^\circ$$

$$\bar{Y}_{23} = \bar{Y}_{32} = -\frac{1}{jx_{23}} = \frac{1}{j0,09917} = j10,0833 \angle 90^\circ$$

$$\bar{Y}_{33} = \frac{1}{jx_{23}} + \frac{jb_{23}}{2} = -j10,0833 + j0,10454 =$$

$$= -j9,9788 = 9,9788 \angle -90^\circ$$

$\bar{Y}_{31} = 0$ (There is no connection)



$$\begin{pmatrix} \bar{Y}_{21} & \bar{Y}_{22} & \bar{Y}_{23} \\ \bar{Y}_{31} & \bar{Y}_{32} & \bar{Y}_{33} \end{pmatrix} \Rightarrow$$

$$\Rightarrow \begin{pmatrix} 10,1382\angle 90^\circ & 20,0164\angle -90^\circ & 10,0833\angle 90^\circ \\ 0 & 10,0833\angle 90^\circ & 9,9711\angle -90^\circ \end{pmatrix}$$

Given Powers:

$P_2 = -\frac{150}{100} = -1,5$, $Q_2 = \frac{-20}{100} = -0,2$

$P_3 = -\frac{95}{100} = -0,95$ $Q_3 = \frac{-9}{100} = -0,09$

$V_1 = \frac{230}{22} = 1,04545$ p.u, $\delta_1 = 0^\circ$

$V_2^{(0)} = V_3^{(0)} = V_1 = 1,04545$ $\delta_2^{(0)} = \delta_3^{(0)} = 0^\circ$

Because Slack Start.

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Decoupled Newton Method:

$$P_i = \sum_j V_i V_j Y_{ij} \cos(\delta_i - \delta_j - \alpha_{ij})$$

$$Q_i = \sum_j V_i V_j Y_{ij} \sin(\delta_i - \delta_j - \alpha_{ij})$$

Full Newton Equation:

$$\begin{pmatrix} \Delta P \\ \Delta Q \end{pmatrix} = \begin{pmatrix} J_1 & J_2 \\ J_3 & J_4 \end{pmatrix} \begin{pmatrix} \Delta \delta \\ \Delta V \end{pmatrix}$$

with

$$J_1 = \frac{\partial P}{\partial \delta} \qquad J_3 = \frac{\partial Q}{\partial \delta}$$

$$J_2 = \frac{\partial P}{\partial V} \qquad J_4 = \frac{\partial Q}{\partial V}$$

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In the decoupled method weak coupling s

$J_2 \approx 0$ and $J_3 \approx 0$ are neglected.

Then we can solve smaller System:

$$\Delta P = J_1 \Delta \delta \qquad \Delta Q = J_4 \Delta V$$

Iteration 1: $V_2^{(0)} = 1{,}04545$ $V_3^{(0)} = 1{,}04545$ $\delta_2^{(0)} = 0$ $\delta_3^{(0)} = 0$

① $P_2^{(0)} = V_2 V_1 Y_{21} \cos(\delta_2 - \delta_1 - \alpha_{21}) + V_2^2 Y_{22} \cos \alpha_{22} +$

$V_2 V_3 Y_{23} \cos(\delta_2 - \delta_3 - \alpha_{23}) = 1{,}04545 \times 1{,}04545 \times$

$\times 10{,}1382 \times \cos(0-0-90) + 1{,}04545^2 \times 20{,}0164 \times \cos(-90) +$

$+ 1{,}04545 \times 1{,}04545 \times 10{,}0833 \times \cos(0-0-90^\circ) = 0$

② $P_3^{(0)} = V_3 V_2 Y_{32} \cos(\delta_3 - \delta_2 - \alpha_{32}) + V_3^2 Y_{33} \cos \alpha_{33} =$

$= 1{,}0455 \times 1{,}0455 \times 10{,}0833 \times \cos(0-0-90) + 1{,}0455^2 \times 9{,}9788 \times$

$\cos(-90) = 0$

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③ $Q_2^{(0)} = V_2 V_1 Y_{21} \operatorname{sen}(\delta_2 - \delta_1 - \alpha_{21}) - V_2^2 Y_{22} \operatorname{sen}\alpha_{22} +$

$V_2 V_3 Y_{23} \operatorname{sen}(\delta_2 - \delta_3 - \alpha_{23}) = 1{,}0455 \times 1{,}0455 \times$

$\times 10{,}1382 \times \operatorname{sen}(0 - 0 - 90°) - 1{,}0455^2 \times 20{,}0964 \times \operatorname{sen}(-90°) +$

$+ 1{,}0455 \times 1{,}0455 \times 10{,}0833 \times \operatorname{sen}(0 - 0 - 90°) =$

$= (-11{,}0809) + (21{,}8774) + (-11{,}0208) = -0{,}2242$

④ $Q_3^{(0)} = V_3 V_2 Y_{32} \operatorname{sen}(\delta_3 - \delta_2 - \alpha_{32}) - V_3^2 Y_{33} \operatorname{sen}\alpha_{33} =$

$= 1{,}0455 \times 1{,}0455 \times 10{,}0833 \times \operatorname{sen}(-90°) - 1{,}0455^2 \times 9{,}9788 \times$

$\times \operatorname{sen}(-90°) = (-11{,}0208) + (10{,}9066) = -0{,}1143$

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Power residuals:

$$\Delta P_2^{(0)} = P_2 - P_2^{(0)} = -1,50 - (0,00) = -1,50$$

$$\Delta P_3^{(0)} = P_3 - P_2^{(0)} = -0,95 - (0,0) = -0,95$$

$$\Delta Q_2^{(0)} = Q_2 - Q_2^{(0)} = -0,20 - (-0,2242) = 0,0242$$

$$\Delta Q_3^{(0)} = Q_3 - Q_3^{(0)} = -0,09 - (-0,1143) = 0,0243$$

Jacobian sub-matrices:

$$J_1 = \frac{\partial P}{\partial \delta} \quad \text{and} \quad J_4 = \frac{\partial Q}{\partial V}$$

① $$\frac{\partial P_2}{\partial \delta_2} = -V_2 V_1 Y_{21} \sin\left(\delta_2 - \delta_1 - \alpha_{21}\right) - V_2 V_3 Y_{23} \sin\left(\delta_2 - \delta_3 - \alpha_{23}\right) =$$

$$= 22,1017$$

② $$\frac{\partial P_2}{\partial \delta_3} = V_2 V_3 Y_{23} \sin\left(\delta_2 - \delta_3 - \alpha_{23}\right) = -11,0208$$

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$$\frac{\partial P_3}{\partial \delta_2} = V_3 V_2 Y_{32} \sin\left(\delta_3 - \delta_2 - \alpha_{32}\right) = -11,0208$$

$$\frac{\partial P_3}{\partial \delta_3} = -V_3 V_2 Y_{32} \sin\left(\delta_3 - \delta_2 - \alpha_{32}\right) = 11,0208$$

$$\frac{\partial Q_2}{\partial V_2} = V_1 Y_{21} \sin\left(\delta_2 - \delta_1 - \alpha_{21}\right) - 2V_2 Y_{22} \sin\alpha_{22} +$$

$$+ V_3 Y_{32} \sin\left(\delta_2 - \delta_3 - \alpha_{23}\right) = 20,7118$$

$$\frac{\partial Q_2}{\partial V_3} = V_2 Y_{23} \sin\left(\delta_2 - \delta_3 - \alpha_{23}\right) = -10,5417$$

$$\frac{\partial Q_3}{\partial V_2} = V_3 Y_{32} \sin\left(\delta_3 - \delta_2 - \alpha_{23}\right) = -10,5417$$

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$$\frac{\partial Q_3}{\partial V_3} = V_2 Y_{32} \sin(\delta_3 - \delta_2 - \alpha_{32}) - 2 V_3 Y_{33} \sin \alpha_{33} =$$

$$= 10{,}3232$$

Angles:

$$\Delta P = J_1 \Delta \delta$$

$$\begin{pmatrix} -1{,}500 \\ -0{,}95 \end{pmatrix} = \begin{pmatrix} 22{,}1017 & -11{,}0208 \\ -11{,}0208 & 11{,}0208 \end{pmatrix} \begin{pmatrix} \Delta \delta_2^{(0)} \\ \Delta \delta_3^{(0)} \end{pmatrix}$$

$$\begin{pmatrix} \Delta \delta_2^{(0)} \\ \Delta \delta_3^{(0)} \end{pmatrix} = J_1^{-1} \Delta P \Rightarrow$$

$$\det J_1 = 22{,}1017 \times 11{,}0208 - (-11{,}0208) \times (-11{,}0208) =$$

$$122{,}1203$$

$$J_1^{-1} = \frac{1}{122{,}1203} \begin{pmatrix} 11{,}0208 & 11{,}0208 \\ 11{,}0208 & 22{,}1017 \end{pmatrix} =$$

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$$J_1^{-1} = \begin{pmatrix} 0,09025 & 0,09025 \\ 0,09025 & 0,18098 \end{pmatrix}$$

$$\begin{pmatrix} \Delta\delta_2^{(0)} \\ \Delta\delta_3^{(0)} \end{pmatrix} = J_1^{-1}\,\Delta P = \begin{pmatrix} 0,09025 & 0,09025 \\ 0,09025 & 0,09025 \end{pmatrix} \begin{pmatrix} -1,50 \\ -0,95 \end{pmatrix}$$

$$= \begin{pmatrix} -0,2211 \\ -0,3073 \end{pmatrix} \text{rad}$$

$$\Delta\delta_2^{(0)} = -0,2211 \times \frac{180^\circ}{\pi} = -12,668^\circ$$

$$\Delta\delta_3^{(0)} = -0,30730 \times \frac{180}{\pi} = -17,607^\circ$$

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Voltage Magnitudes:

$$\Delta Q = J_4 \Delta V$$

$$\begin{pmatrix} 0,0242 \\ 0,0243 \end{pmatrix} = \begin{pmatrix} 20,7118 & -10,5417 \\ -10,5417 & 10,3231 \end{pmatrix} \begin{pmatrix} \Delta V_2^{(0)} \\ \Delta V_3^{(0)} \end{pmatrix}$$

$$J_4^{-1} = \begin{pmatrix} 0,10053 & 0,10266 \\ 0,10266 & 0,20171 \end{pmatrix}$$

$$\begin{pmatrix} \Delta V_2^{(0)} \\ \Delta V_3^{(0)} \end{pmatrix} = J_4^{-1} \Delta Q = \begin{pmatrix} 0,10053 & 0,10266 \\ 0,10266 & 0,20171 \end{pmatrix} \begin{pmatrix} 0,0242 \\ 0,0243 \end{pmatrix}$$

$$= \begin{pmatrix} 0,00493 \\ 0,00733 \end{pmatrix} \text{p.u}$$

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Update of bus Voltages:

$V_2^{(1)} = V_2^{(0)} + \Delta V_2^{(0)} = 1{,}0455 + 0{,}00491 = 1{,}0504$ p.u.

$V_3^{(1)} = V_3^{(0)} + \Delta V_3^{(0)} = 1{,}0455 + 0{,}00728 = 1{,}0528$

$\delta_2^{(1)} = \delta_2^{(0)} + \Delta \delta_2^{(0)} = -12{,}668°$

$\delta_3^{(1)} = \delta_3^{(0)} + \Delta \delta_3^{(0)} = -17{,}607$

$V_2^{(1)} = 235{,}08\,kV \qquad V_3^{(1)} = 231{,}62\,kV$

Second Iteration:

Starting Values →

$V_2^{(1)} = 1{,}0504 \quad V_3^{(1)} = 1{,}0528 \quad \delta_2^{(1)} = -12{,}668°$

$\delta_3^{(1)} = -17{,}607°$

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[Powers:]

$$P_2^{(1)} = V_2 V_1 Y_{21} \cos(\delta_2 - \delta_1 - \alpha_{21}) + V_2^2 Y_{22} \cos\alpha_{22} + V_2 V_3 \cos(\delta_2 - \delta_3 - \alpha_{23}) = -1,4815$$

$$P_3^{(1)} = V_3 V_2 Y_{32} \cos(\delta_3 - \delta_2 - \alpha_{32}) + V_3^2 Y_{33} \cos\alpha_{33} =$$

$$= -0,9600$$

$$Q_2^{(1)} = V_2 V_1 Y_{21} \sin(\delta_2 - \delta_1 - \alpha_{21}) - V_2^2 Y_{22} \sin\alpha_{22} + V_2 V_3 Y_{23} \sin(\delta_2 - \delta_3 - \alpha_{23}) = 0,1125$$

$$Q_3^{(1)} = V_3 V_2 Y_{32} \sin(\delta_3 - \delta_2 - \alpha_{32}) - V_3^2 Y_{33} \sin\delta_{33} = -0,0484$$

$$\Delta P_2^{(1)} = P_2 - P_2^{(1)} = -0,0185$$

$$\Delta P_3^{(1)} = 0,0106$$

$$\Delta Q_2^{(1)} = -0,3125$$

$$\Delta Q_3^{(1)} = -0,0416$$

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Jacobian sub-matrices:

$$J_1 = \frac{\partial P}{\partial \delta} \qquad J_4 = \frac{\partial Q}{\partial V}$$

① $$\frac{\partial P_2}{\partial \delta_2} = -V_2 V_1 Y_{21} \sin(\delta_2 - \gamma_1 - \alpha_{21}) - V_2 V_3 Y_{23} \sin(\delta_2 - \delta_3 - \alpha_{23})$$

$$= 21,9716$$

② $$\frac{\partial P_2}{\partial \delta_3} = V_2 V_3 Y_{23} \sin(\delta_2 - \delta_3 - \alpha_{23}) = -11,1096$$

③ $$\frac{\partial P_3}{\partial \delta_2} = V_3 V_2 Y_{32} \sin(\delta_3 - \delta_2 - \alpha_{32}) = -11,1096$$

④ $$\frac{\partial P_3}{\partial \delta_3} = -V_3 V_2 Y_{32} \sin(\delta_3 - \delta_2 - \alpha_{32}) = 11,1096$$

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⑤ $\frac{\partial Q_2}{\partial V_2} = V_1 Y_{21} \sin(\delta_2 - \delta_1 - \alpha_{21}) - 2V_2 Y_{22} \sin\alpha_{22} +$

$+ V_3 Y_{23} \sin(\delta_2 - \delta_3 - \alpha_{23}) = 21,1320$

⑥ $\frac{\partial Q_3}{\partial V_2} = V_3 Y_{32} \sin(\delta_3 - \delta_2 - \alpha_{32}) = -10,5767$

⑦ $\frac{\partial Q_3}{\partial V_3} = V_2 Y_{32} \sin(\delta_3 - \delta_2 - \alpha_{32}) - 2V_3 Y_{33} \sin\alpha_{33} =$

$= 10,1601$

⑧ $\frac{\partial Q_2}{\partial V_3} = V_2 Y_{23} \sin(\delta_2 - \delta_3 - \alpha_{23}) = -10,5520$

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Angles:

$$\Delta P = J_1 \Delta \delta$$

$$\begin{pmatrix} \Delta \delta_2^{(1)} \\ \Delta \delta_3^{(1)} \end{pmatrix} = J_1^{-1} \Delta P = \begin{pmatrix} 909206 & 0{,}09206 \\ 0{,}09206 & 0{,}18208 \end{pmatrix} \begin{pmatrix} -0{,}0185 \\ 0{,}0100 \end{pmatrix}$$

$$= \begin{pmatrix} -0{,}00078 \\ 0{,}000112 \end{pmatrix} \text{rad}$$

$$\Delta \delta_2^{(1)} = -0{,}00078 \times \frac{180}{\pi} = -0{,}045°$$

$$\Delta \delta_3^{(1)} = 0{,}000112 \times \frac{180}{\pi} = 0{,}007°$$

Voltage magnitudes:

$$\Delta Q_1 = J_4 \Delta V$$

$$\begin{pmatrix} -0{,}3125 \\ -0{,}0416 \end{pmatrix} = \begin{pmatrix} 21{,}1320 & -10{,}5520 \\ -10{,}5767 & -20{,}4601 \end{pmatrix} \begin{pmatrix} \Delta V_2^{(1)} \\ \Delta V_3^{(1)} \end{pmatrix}$$

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$$\begin{pmatrix} \Delta V_2^{(1)} \\ \Delta V_3^{(1)} \end{pmatrix} = J^{-1} \Delta Q = \begin{pmatrix} 0,09556 & 0,09642 \\ 0,09665 & 0,19310 \end{pmatrix} \begin{pmatrix} -0,3125 \\ -0,0416 \end{pmatrix}$$

$$= \begin{pmatrix} -0,033386 \\ -0,033824 \end{pmatrix} \text{ p.u.}$$

Update of bus Voltages:

$V_2^{(2)} = V_2^{(1)} + \Delta V_2^{(1)} = 1,0165$ p.u.

$V_3^{(2)} = V_3^{(1)} + \Delta V_3^{(1)} = 1,0146$ p.u.

$\delta_2^{(2)} = \delta_2^{(1)} + \Delta \delta_2^{(1)} = -12,713°$

$\delta_3^{(2)} = \delta_3^{(1)} + \Delta \delta_3^{(1)} = -17,60°$

Result: $V_3^{(2)} = 1,0146$ p.u. $\quad V_3^{(2)} = 223,11$ kV

$\left[V_3^{(2)} = 223,11\text{kV} \angle -17,60° \right]$

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